Life cycle of SQL Query Execution

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From Java application if we submit SQL query by using Statement object execute method,

```
Statement st = con.createStatement();
ResultSet rs = st.executeQuery(sqlQuery);
```

Then database engine will perform the following sequence of activities:

a. Compilation
b. Execution
c. Fetch result

a. Compilation: As a part of compilation, database engine will perform the following activities:

i. **Query Tokenization:** In this step total SQL query will be divided into number of tokens and generate a stream of tokens as output.

ii. **Query Parsing:** In this step database engine will create parse tree (Queue tree) with stream of tokens. If the query tree is proper, then there are no syntactical mistake in that query. If query tree construction fails then it indicates that there are some syntactical error present in SQL query and SQLException will be raised.

iii. **Query Optimization:** The main purpose of Optimization is to improve performance. In this step optimized query tree will be created.

b. Execution of SQL Query: Once compilation success then database engine will take the query tree as input and execute that query by using interpreter.

c. Fetch the result: Database engine will provide result of SQL query either in the form of ResultSet (for select query) or in the form of row count (for non-select query) to the Java application.